

Grouping 1

Figure 2 (Phase V Sweep Width)

Generated From

Experimental BCTDS measurements (I/Q quadrature time-series) recorded from the broadband waveguide experiment

Sensors?

AMD rfsoc 4x2 fpga, using QICK package

Experiment Set Up

What temperature, power, any pre-set up before the experiment was applied?

10 milik, sets pulse width each time,

```
def convert_clock_ticks(input_key, input_value, output_key):
    clock_freqs_MHz = {"time_us": 1, "sig_gen": 9830.4, "t_proc": 409.6, "decimated_RO":
552.96}
    if input_key not in clock_freqs_MHz or output_key not in clock_freqs_MHz:
        raise ValueError("Invalid key. Must be one of: " + ", ".join(clock_freqs_MHz.keys()))
    return input_value / clock_freqs_MHz[input_key] * clock_freqs_MHz[output_key]
```

How long was the measurement of data?

~1.8 ms across the 1000

~0.8 ms across the 500

You can get the time in ms by doing 0/552.96

Data in Folder (next page)

File Name(s)	Description	# of Files	Shape	Features	Size
*_IQ_avg_matrix_0.npy	Goes 50-2100, ex. 50,100,150 etc-> Avg -> instrument level avg, uses buffer onboard for software averages, AMD device 50 -> pulse length (how long the pulse is on) /~9.8 to get it to real time ns, this is the devices' clock time	42	(2, 500, 1000)	(2):[['I matrix for time versus frequency'], ['Q matrix for time versus frequency']] (500): 500 mghz span, every mghz you collect (1000): Time Points, 552.96 spacing -> to actual spacing in microseconds	1000000

File Name	Description	# of Files	Shape	Features	Size
full_record_log_mag.npy	Guessing this is all of the previous 42 files in a different format?	1	(42, 500, 835)	(2): Previous 42? (500): 500 mghz span, every mghz you collect (835): Accounting for gap that is before the pulse	17535000 Why is this different?

File Name	Description	# of Files	Shape	Features	Size
full_record_phase_fft.npy	Guessing this is all of the previous 42 files in a different format?	1	(42, 500, 417)	(2): Previous 42? (500): Is this different? (417): Time Steps? Why is this now 417?	8757000 Why is this different?

Fft -> frequency components that exists

File Name	Description	# of Files	Shape	Features	Size
peak_plot_data.npz	Unknown, where is this data coming from?	1	Freq_axis: (500,) Y: (500,) display_freq_range: (2,)	Contains 3 array(s): ['freq_axis', 'y', 'display_freq_range']	3 Arrays 500 500 2

File Name	Description	# of Files	Shape	Features	Size
phase_fft_slice_and_avg.npz	Unknown, where is this data coming from?	1	9 Arrays Freq_axis 500 y_axis 42 display_freq_range 2 phase_fft_slice (42, 500) imshow_cmap 1 (what are the imshows?) imshow_origin 1 imshow_vmin 1 imshow_vmax 1 avg_line 500	phase_fft_slice unknown for both features of the array	phase_fft_slice 21000

File Name	Description	# of Files	Shape	Features	Size
slice_0_at_diff_widths.npy	Unknown, where is this data coming from?	1	(42,500)	Unknown Names	21000

File Name	Description	# of Files	Shape	Features	Size
slice_0_avg_at_diff_widths.npy	Unknown, where is this data coming from?	1	(42,500)	Unknown Names	21000

File Name	Description	# of Files	Shape	Features	Size
slice_later_at_diff_widths.npy	Unknown, where is this data coming from?	1	(42,500)	Unknown Names	21000

File Name	Description	# of Files	Shape	Features	Size
slice_later_avg_at_diff_widths.npy	Unknown, where is this data coming from?	1	(42,500)	Unknown Names	21000

Figure 3 (Coherent Control Amplitude)

Generated From

Experimental BCTDS measurements (I/Q quadrature time-series) recorded from the broadband waveguide experiment

Experiment Set Up

What temperature, power, any pre-set up before the experiment was applied?

Unknown

How long was the measurement of data?

I believe we had said 1.6 seconds after turning on and then off?

And is this reflected in the data in some way?

Is this the same for every piece of data within this folder?

Data in Folder

File Name(s)	Description	# of Files	Shape	Features	Size
111_oxide_308_freq_4254_spacing_***_IQ_avg_matrix.npy	*** goes 0.0-0.05 What are the naming conventions for? 111? Oxide? 308? Freq_4254? Spacing 0.0-0.05? Avg?	6	(2, 100, 1000)	(2): What are these two features? (100): Guessing this is how many times the two features are captured? (1000): Time Steps? Does this indicate anything else	200000

File Name(s)	Description	# of Files	Shape	Features	Size
111_oxide_HFSS_calibrated_3-5_GHz_IQ_avg_matrix.npy	What are the naming conventions for? Need more context	1	(2, 2000, 1000)	(2): What are these two features? (2000): Guessing this is how many times the two features are captured? (1000): Time Steps? Does this indicate anything else	4000000

File Name(s)	Description	# of Files	Shape	Features	Size
AlOx_308_freq_3657_spacing_*****_IQ_avg_matrix.npy	*** goes 0.0-0.05 What are the naming conventions for? AlOx? 308? Freq_3657? Spacing 0.0-0.05? Avg?	6	(2, 100, 1000)	(2): What are these two features? (100): Guessing this is how many times the two features are captured? (1000): Time Steps? Does this indicate anything else	200000

File Name(s)	Description	# of Files	Shape	Features	Size
AlOx_HFSS_calibrated_3-5_GHz_IQ_avg_matrix.npy	What are the naming conventions for? Need more context	1	(2, 2000, 1000)	(2): What are these two features? (2000): Guessing this is how many times the two features are captured? (1000): Time Steps? Does this indicate anything else	4000000

File Name(s)	Description	# of Files	Shape	Features	Size
silicon_308_freq_4424_spacing_****_IQ_avg_matrix.npy	*** goes 0.0-0.05 What are the naming conventions for? Silicon? 308? Freq_4424? Spacing 0.0-0.05? Avg?	6	(2, 100, 1000)	(2): What are these two features? (100): Guessing this is how many times the two features are captured? (1000): Time Steps? Does this indicate anything else	200000

File Name(s)	Description	# of Files	Shape	Features	Size
silicon_HFSS_calibrated_3-5_GHz_IQ_avg_matrix.npy	What are the naming conventions for? Need more context	1	(2, 2000, 1000)	(2): What are these two features? (2000): Guessing this is how many times the two features are captured? (1000): Time Steps? Does this indicate anything else	4000000

Figure 4 (Coherent Control Phase)

Generated From

Experimental BCTDS measurements (I/Q quadrature time-series) recorded from the broadband waveguide experiment

Experiment Set Up

What temperature, power, any pre-set up before the experiment was applied?

Unknown

How long was the measurement of data?

I believe we had said 1.6 seconds after turning on and then off?

And is this reflected in the data in some way?

Is this the same for every piece of data within this folder?

Data in Folder

So I'm seeing same naming conventions as Figure 3, what is the overall difference here in the data?

Are they correlated in any way? Are they from the same run?

Figure 5 (Coherent Control Phase Long Spacing)

Generated From

Experimental BCTDS measurements (I/Q quadrature time-series) recorded from the broadband waveguide experiment

Experiment Set Up

What temperature, power, any pre-set up before the experiment was applied?

Unknown

How long was the measurement of data?

I believe we had said 1.6 seconds after turning on and then off?

And is this reflected in the data in some way?

Is this the same for every piece of data within this folder?

Data in Folder

File Name(s)	Description	# of Files	Shape	Features	Size
308_freq_3400_spacing_****_ID_Shipley_phase_8mK_20251012_132853.npz	*** goes 0.01, 0.03, 0.05, then 0.1-0.6 What are the naming conventions for? 308? Freq_3400? Shipley? 8mK_20251012_132853? Spacing values new 0.01,0.03,0.05 and then 0.1-0.6?	9	Contains 3 array(s): ['IQ_avg_matrix', 'pulse_phase_array', 'time_stamp_list'] IQ_avg_matrix Shape: (2, 121, 1000) pulse_phase_array Shape: (121,) Why do these go up by 3's? 0 3 6 9... time_stamp_list Shape: (121,) 2025-10-12T14:44:30	(2): What are these two features? (121): Guessing this is how many times the two features are captured? (1000): Time Steps? Does this indicate anything else	242000

Figure 14 (Coherent Control Phase Long Spacing)

Generated From

Experimental BCTDS measurements (I/Q quadrature time-series) recorded from the broadband waveguide experiment

Experiment Set Up

What temperature, power, any pre-set up before the experiment was applied?

Unknown

How long was the measurement of data?

I believe we had said 1.6 seconds after turning on and then off?

And is this reflected in the data in some way?

Is this the same for every piece of data within this folder?

Data in Folder

File Name(s)	Description	# of Files	Shape	Features	Size
1950_amp_****_ IQ_avg_matrix_ 0.npy	*** goes 0-1000 by 50s Then 1000-5000 by 200 Then 5000-30000 by 1000 Why 1950?	66	(2, 500, 1000)	(2): What are these two features? (500): Guessing this is how many times the two features are captured? (1000): Time Steps? Does this indicate anything else	1000000

File Name(s)	Description	# of Files	Shape	Features	Size
complete_pulse_ amp_record.n py	Is there anything unique to this, or is it just the full 66 all together?	1	(66, 500, 417)	(66): Are these the previous 66? (500): Guessing this is how many times the two features are captured? (417): Time Steps? Does this indicate anything else	13761000

File Name(s)	Description	# of Files	Shape	Features	Size
full_record_log_mag.npy	Is there anything unique to this, or is it just the full 66 all together? What is the difference between this and the previous?	1	(66, 500, 835)	(66): Are these the previous 66? (500): Guessing this is how many times the two features are captured? (835): Time Steps? Does this indicate anything else	27555000

File Name	Description	# of Files	Shape	Features	Size
slice_0_at_diff_amps.npy	Unknown, where is this data coming from?	1	(66,500)	Unknown Names	33000

File Name	Description	# of Files	Shape	Features	Size
slice_0_avg_at_diff_widths.npy	Unknown, where is this data coming from?	1	(66,500)	Unknown Names	33000

File Name	Description	# of Files	Shape	Features	Size
slice_later_at_diff_amps.npy	Unknown, where is this data coming from?	1	(66,500)	Unknown Names	33000

File Name	Description	# of Files	Shape	Features	Size
slice_later_avg_at_diff_amps.npy	Unknown, where is this data coming from?	1	(66,500)	Unknown Names	33000

Grouping 2

Figure 11 (temperature_sweep_during_warmup)

Generated From

Temperature-dependent BCTDS experiments during dilution refrigerator warm-up and thermal cycling. Guessing these experiments probe how TLS spectral structure and coherence evolve with temperature.

Data likely originates from:

- cryogenic waveguide measurements
- silicon or sapphire samples with oxide layers
- homodyne detection producing I/Q signals

What the figures are showing:

- TLS spectral responses across different temperatures
- stability or reconfiguration of TLS eigenfrequencies
- thermal dependence of ring-down dynamics
- changes in interference patterns under coherent control
- evidence that TLS spectral locations can change after thermal cycling

Experiment Set Up

Paper documents the process, what's the one sentence breakdown?

File Name	Description	# of Files	Shape	Features	Size
308_rep_***_ID_FM_Shipley_cont_HFSS_calib_20251008_174837_with_T.npz	0-105? What is this?	106	Contains 5 array(s): ['IQ_avg_matrix', 'pulse_freq_array', 'time_stamp_list', 'T_still', 'T_mxc'] Shape: (2, 401, 1000) Remaining shapes: Shape: (401,)		802000

File Name	Description	# of Files	Shape	Features	Size
308_rep_*.ID_FM_Shipley_cont_HFSS_calib_20251008_174837_wit_h_T.npz	***0-105? What is this Contains 13 array(s): ['freq_axis', 'mag_log_matrix', 'time_axis', 'tau_us_array', 'xc_array', 'average_transmission', 'time_stamp_list', 'T_still', 'T_mxc', 'phase_TP_fft_matrix', 'interpolated_freq_axis', 'interpolated_fft_freq_axis', 'V_fit_avg_phase_TP_fft_matrices']	106	Shape: (401,) Shape: (401, 910) Shape: (910,) Shape: (401,) Shape: (401,) Shape: (401,) Shape: (401,) Shape: (401,) Shape: (401,) Array: phase_TP_fft_matrix Shape: (2001, 276) Array: interpolated_freq_axis --- Shape: (2001,) Array: interpolated_fft_freq_axis --- Shape: (276,) Array: V_fit_avg_phase_TP_fft_matrices --- Shape: (2001,)	Need explanations here	Straight forward

Figure 12 (coherent_control_phase_diff_T)

Generated From

Temperature-dependent BCTDS experiments during dilution refrigerator warm-up and thermal cycling. Guessing these experiments probe how TLS spectral structure and coherence evolve with temperature.

Data likely originates from:

- cryogenic waveguide measurements
- silicon or sapphire samples with oxide layers
- homodyne detection producing I/Q signals

What the figures are showing:

- TLS spectral responses across different temperatures
- stability or reconfiguration of TLS eigenfrequencies
- thermal dependence of ring-down dynamics
- changes in interference patterns under coherent control
- evidence that TLS spectral locations can change after thermal cycling

Experiment Set Up

Figure 13 (thermal_cycle)

Generated From

Temperature-dependent BCTDS experiments during dilution refrigerator warm-up and thermal cycling. Guessing these experiments probe how TLS spectral structure and coherence evolve with temperature.

Data likely originates from:

- cryogenic waveguide measurements
- silicon or sapphire samples with oxide layers
- homodyne detection producing I/Q signals

What the figures are showing:

- TLS spectral responses across different temperatures
- stability or reconfiguration of TLS eigenfrequencies
- thermal dependence of ring-down dynamics
- changes in interference patterns under coherent control
- evidence that TLS spectral locations can change after thermal cycling

Experiment Set Up

- Figure 11 (Experiment 11) : What is the philosophy behind this experiment?
- processed_result_npz
 - Processed lifetime - explain
 - Feature 1
 - Etc
- npz_data_base_with_T - Give me an explanation here?
 - 308 blablablah - Explanation
 - Feature 1
 - Feature 2
 - Feature 3